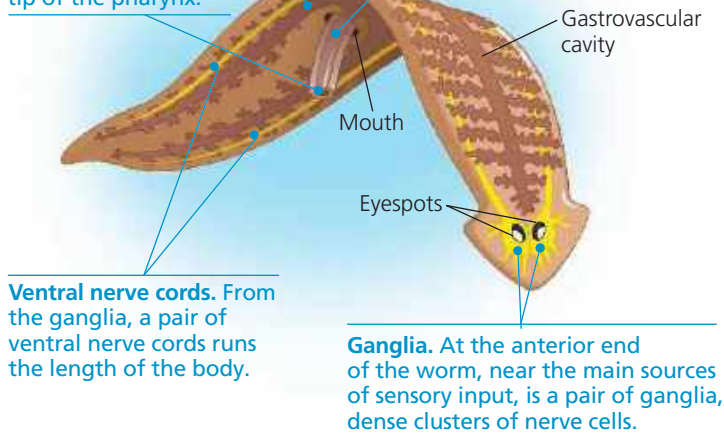


▼ **Figure 33.9 Anatomy of a planarian.**

Digestion is completed within the cells lining the gastrovascular cavity, which has many fine subbranches that provide an extensive surface area.

Undigested wastes are eliminated through an opening at the tip of the pharynx.

Pharynx. A muscular pharynx can be extended through the mouth. Digestive juices are spilled onto prey, and the pharynx sucks small pieces of food into the gastrovascular cavity, where digestion continues.



(Figure 33.9). Experiments have shown that planarians can learn to modify their responses to stimuli.

Some planarians can reproduce asexually through fission. The parent constricts roughly in the middle of its body, separating into a head end and a tail end; each end then regenerates the missing parts. Sexual reproduction also occurs. Planarians are hermaphrodites, and copulating mates typically cross-fertilize each other.

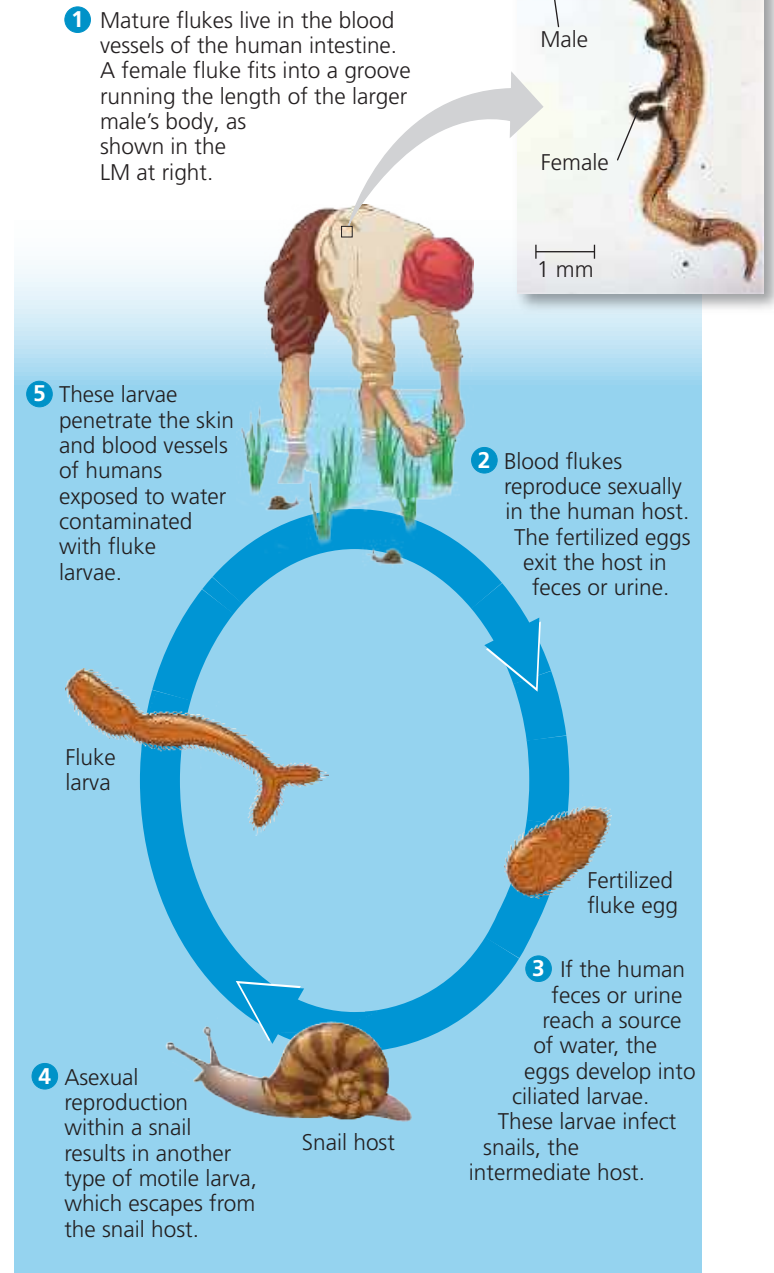
Parasitic Species

More than half of the known species of rhabditophorans live as parasites in or on other animals. Many have suckers that attach to the internal organs or outer surfaces of the host animal. In most species, a tough covering helps protect the parasites within their hosts. We'll discuss two ecologically and economically important subgroups of parasitic rhabditophorans, the trematodes and the tapeworms.

Trematodes As a group, trematodes parasitize a wide range of hosts, and most species have complex life cycles with alternating sexual and asexual stages. Many trematodes require an intermediate host in which larvae develop before infecting the final host (usually a vertebrate), where the adult worms live. For example, various trematodes that parasitize humans spend part of their lives in snail hosts (Figure 33.10). Around the world, about 200 million people are infected with trematodes called blood flukes (*Schistosoma*) and suffer from schistosomiasis, a disease whose symptoms include pain, anemia, and diarrhea.

Living within more than one kind of host puts demands on trematodes that free-living animals don't face. A blood fluke, for instance, must evade the immune systems of both

▼ **Figure 33.10 The life cycle of a blood fluke (*Schistosoma mansoni*), a trematode.**



WHAT IF? Snails eat algae, whose growth is stimulated by nutrients found in fertilizer. How would the contamination of irrigation water with fertilizer likely affect the occurrence of schistosomiasis? Explain.

snails and humans. By mimicking the surface proteins of its hosts, the blood fluke creates a partial immunological camouflage for itself. It also releases molecules that manipulate the hosts' immune systems into tolerating the parasite's existence. These defenses are so effective that individual blood flukes can survive in humans for more than 40 years.

Tapeworms The tapeworms are a second large and diverse group of parasitic rhabditophorans (Figure 33.11). The adults